Design of Enterprise IoT

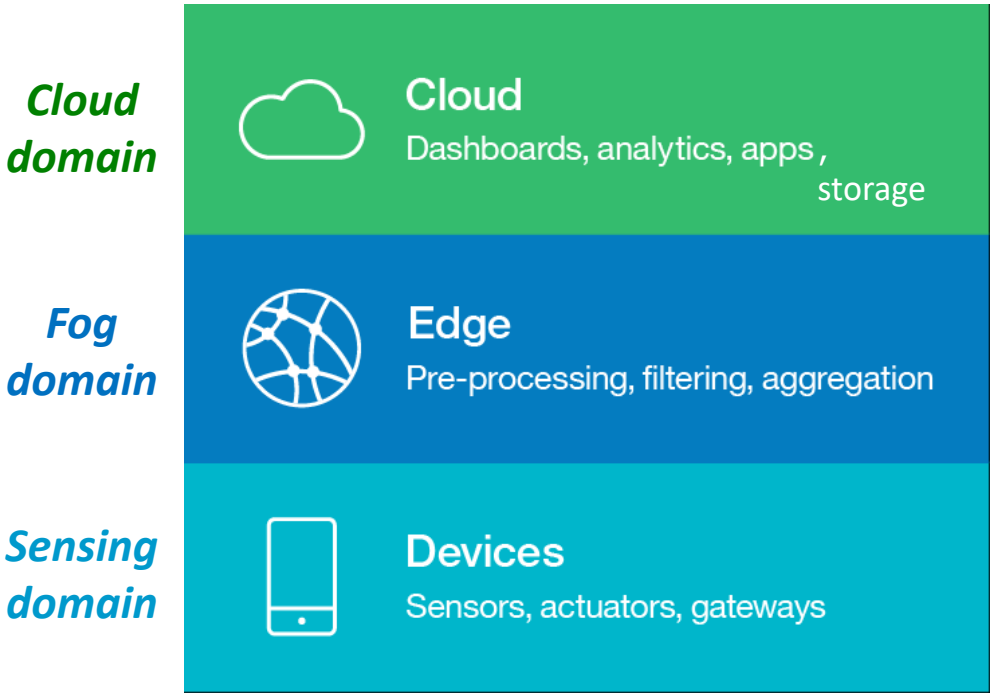
Lecture 4

Cloud & Fog Computing

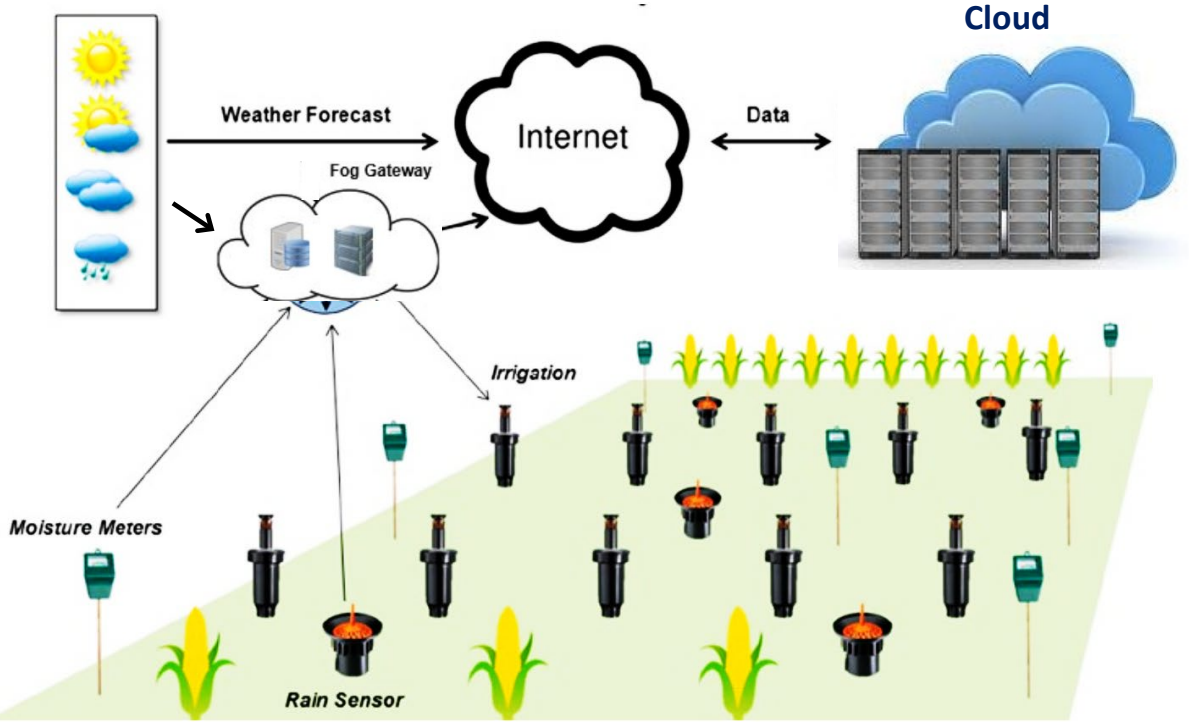
A/Prof Chun Ouyang



Recap: IoT Layered Architecture



Revisit Example: Smart Irrigation



(daCosta & Henderson, 2014)

Cloud Computing

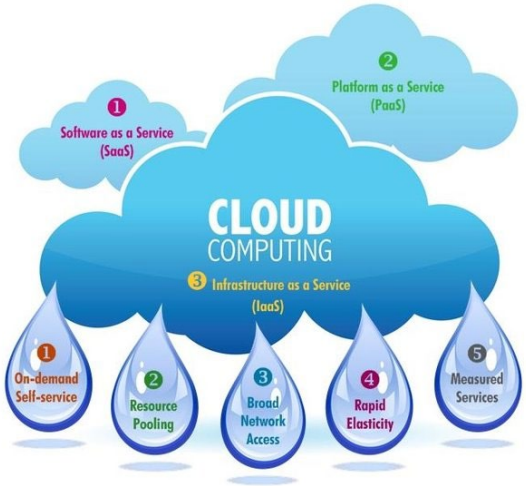


image address

What is Cloud Computing?

- Cloud computing (or simply, “the cloud”) describes the act of storing, managing and processing data online – as opposed to on your physical computer or network.

  Ref: *Rackspace* (https://www.rackspace.com/cloud/cloud-computing)

- Cloud computing is the on-demand delivery of compute power, database storage, applications, and other IT resources through a cloud services platform via the Internet with pay-as-you-go pricing.

  Ref: *AWS* (https://aws.amazon.com/what-is-cloud-computing)

- Cloud computing is the use of computing resources — servers, database management, data storage, networking, software applications, and special capabilities such as blockchain and artificial intelligence (AI) — over the Internet, as opposed to owning and operating those resources yourself, on premises.

  Ref: *IBM* (https://www.ibm.com/cloud/learn/cloud-computing)

What is Cloud Computing?

- Cloud computing (or simply, "the cloud") describes the act of storing, managing and processing data **online** – as opposed to on your physical computer or network.

  Ref: *Rackspace* (https://www.rackspace.com/cloud/cloud-computing)

- Cloud computing is the on-demand delivery of compute power, database storage, applications, and other IT resources through a cloud services platform via the Internet with pay-as-you-go pricing.

  Ref: *AWS* (https://aws.amazon.com/what-is-cloud-computing)

- Cloud computing is the use of computing resources — servers, database management, data storage, networking, software applications, and special capabilities such as blockchain and artificial intelligence (AI) — over the Internet, as opposed to owning and operating those resources yourself, on premises.

  Ref: *IBM* (https://www.ibm.com/cloud/learn/cloud-computing)

What is Cloud Computing?

- Cloud computing (or simply, "the cloud") describes the act of storing, managing and processing data **online** – as opposed to on your physical computer or network.

  Ref: *Rackspace* (https://www.rackspace.com/cloud/cloud-computing)

- Cloud computing is the **on-demand** delivery of compute power, database storage, applications, and other IT resources through a cloud services platform via the Internet with **pay-as-you-go** pricing.

  Ref: *AWS* (https://aws.amazon.com/what-is-cloud-computing)

- Cloud computing is the use of computing resources — servers, database management, data storage, networking, software applications, and special capabilities such as blockchain and artificial intelligence (AI) — over the Internet, as opposed to owning and operating those resources yourself, on premises.

  Ref: *IBM* (https://www.ibm.com/cloud/learn/cloud-computing)

What is Cloud Computing?

- Cloud computing (or simply, "the cloud") describes the act of storing, managing and processing data **online** – as opposed to on your physical computer or network.

  Ref: *Rackspace* (https://www.rackspace.com/cloud/cloud-computing)

- Cloud computing is the **on-demand** delivery of compute power, database storage, applications, and other IT resources through a cloud services platform via the internet with **pay-as-you-go** pricing.

  Ref: *AWS* (https://aws.amazon.com/what-is-cloud-computing)

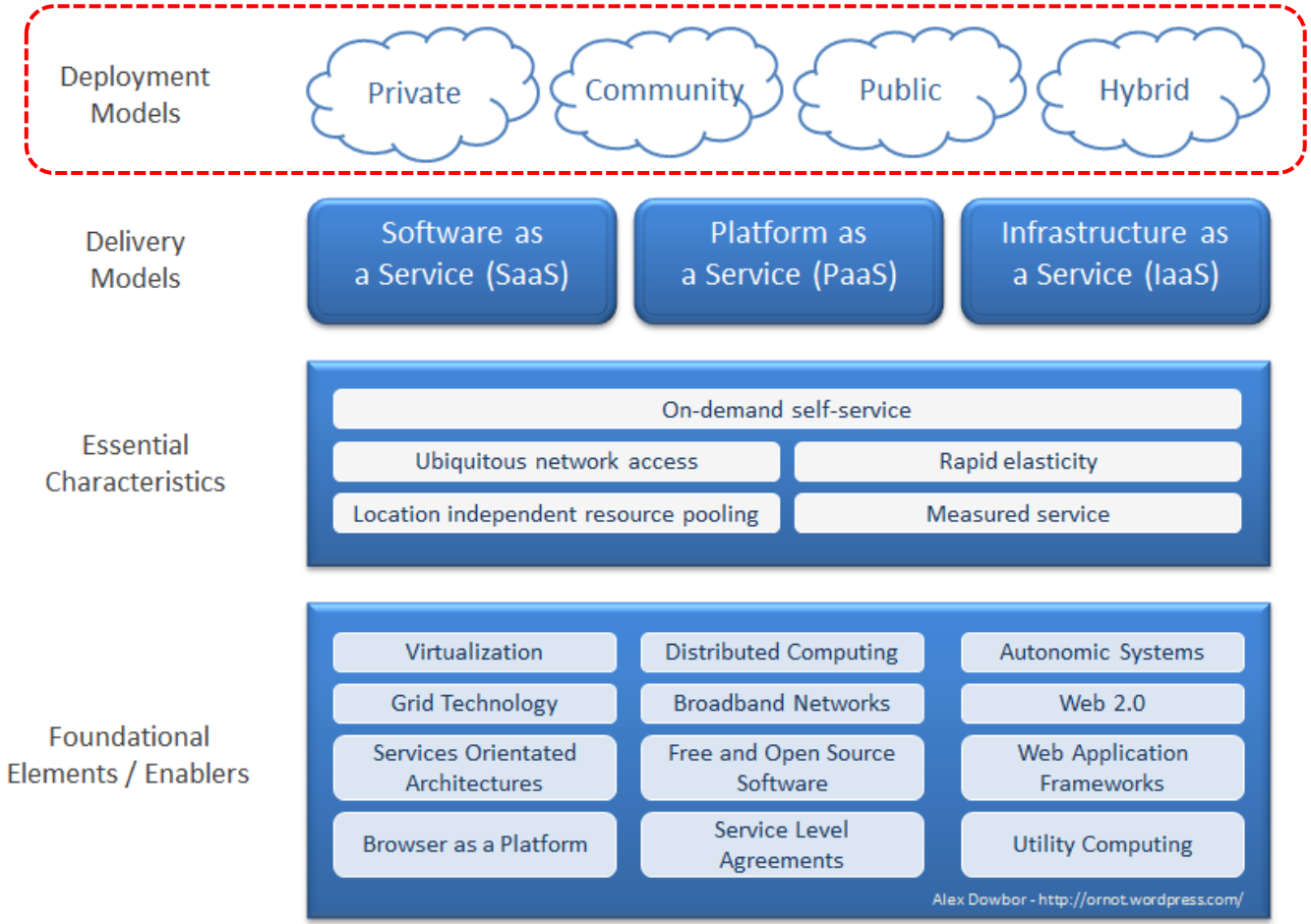
- Cloud computing is the use of **computing resources** — servers, database management, data storage, networking, software applications, and special capabilities such as blockchain and artificial intelligence (AI) — over the Internet, as opposed to owning and operating those resources yourself, on premises.

  Ref: *IBM* (https://www.ibm.com/cloud/learn/cloud-computing)

Definition by NIST

- Cloud computing is a model for enabling ubiquitous, convenient, *on-demand* network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be *rapidly* provisioned and released with *minimal* management effort or service provider interaction.

by National Institute of Standard and Technology (NIST) in USA
https://csrc.nist.gov/publications/detail/sp/800-145/final



Based on the NIST Working Definition of Cloud Computing v14 and http://www.csrc.nist.gov/groups/SNS/cloud-computing/index.html

Creative Commons Attribution-Share Alike 3.0
Alexander Dowbor – http://ornot.wordpress.com

Cloud Services *Deployment Models*

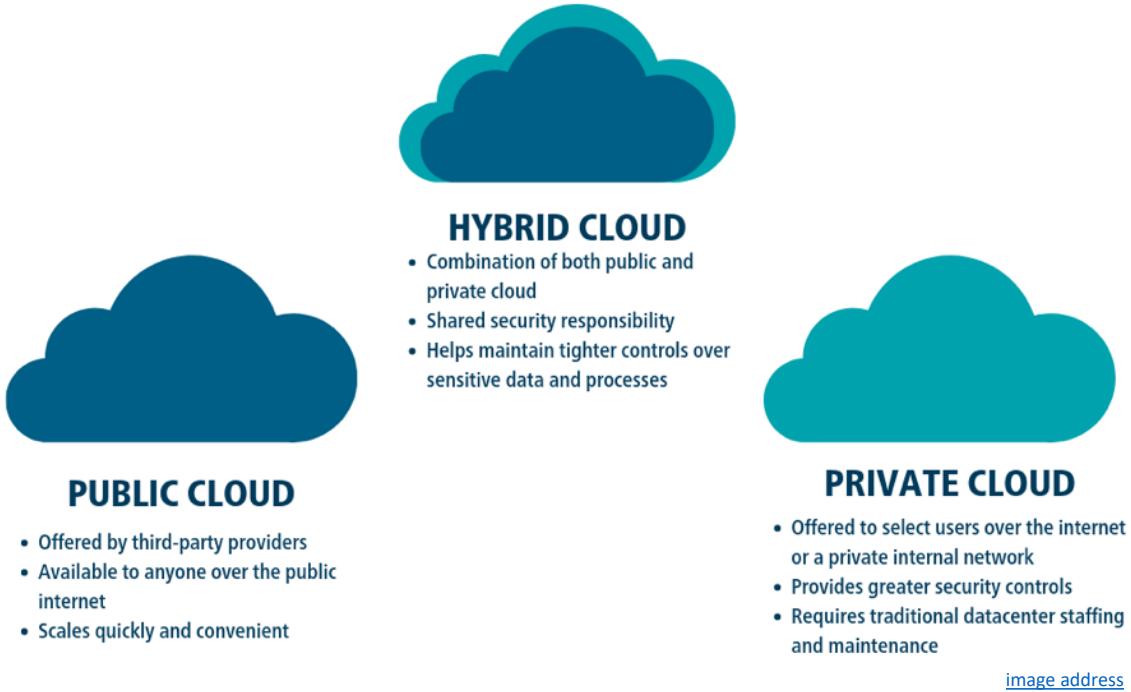


image address

- Five real-life examples of hybrid cloud: https://www.knowledgenile.com/blogs/real-life-examples-hybrid-cloud/

Cloud Services *Deployment Models*

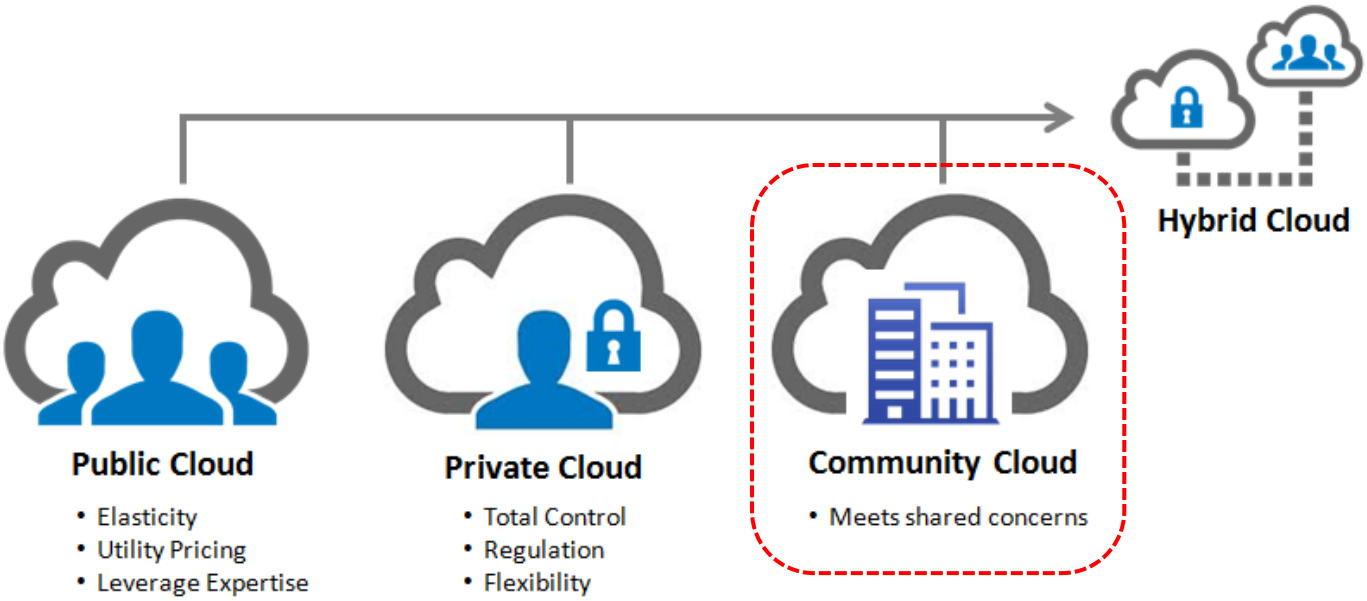
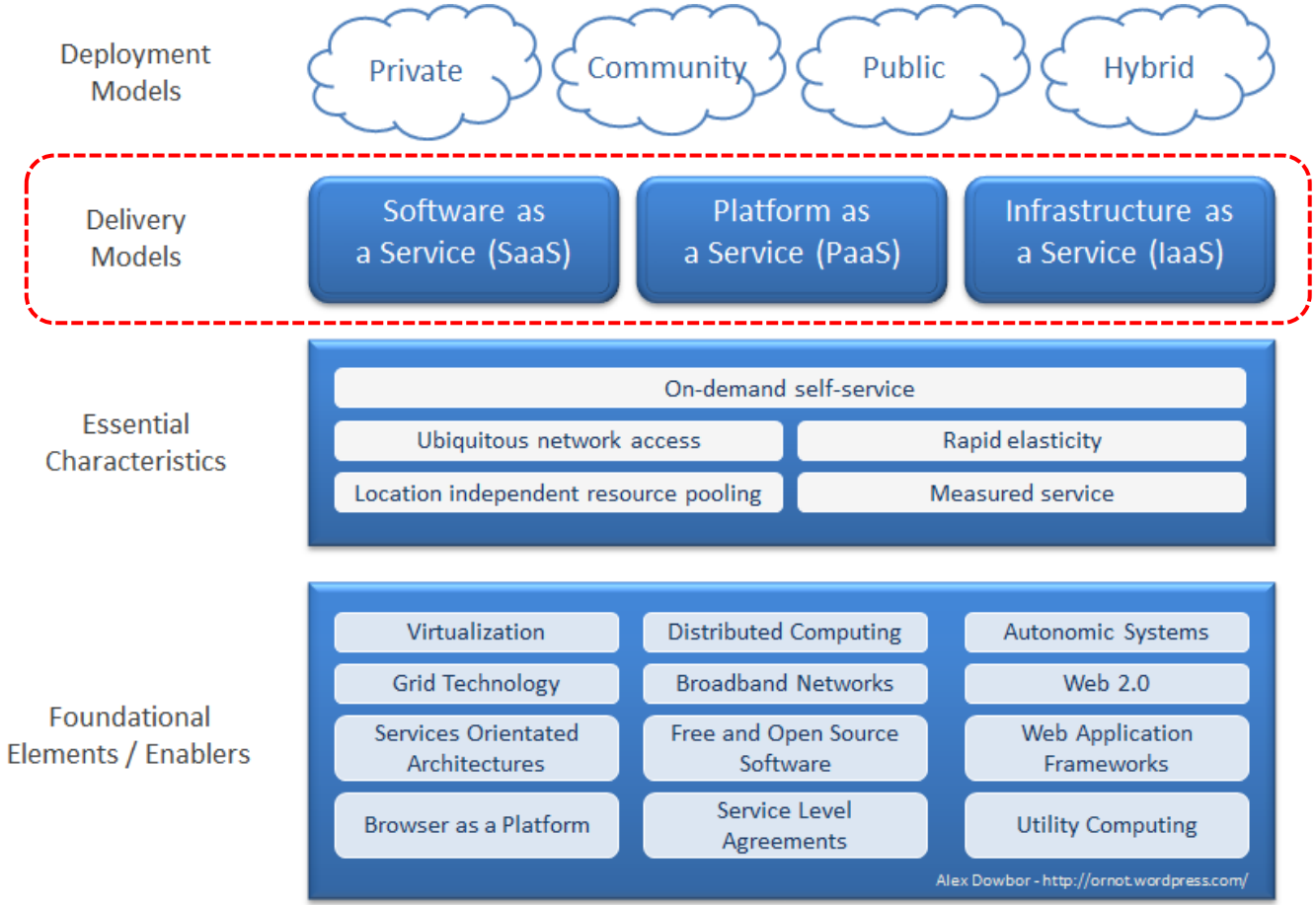


image address

- Salesforce Community Cloud
  https://www.salesforce.com/au/products/community-cloud/overview/



Based on the NIST Working Definition of Cloud Computing v14 and http://www.csrc.nist.gov/groups/SNS/cloud-computing/index.html

Creative Commons Attribution-Share Alike 3.0
Alexander Dowbor – http://ornot.wordpress.com

Cloud Services *Delivery Models*

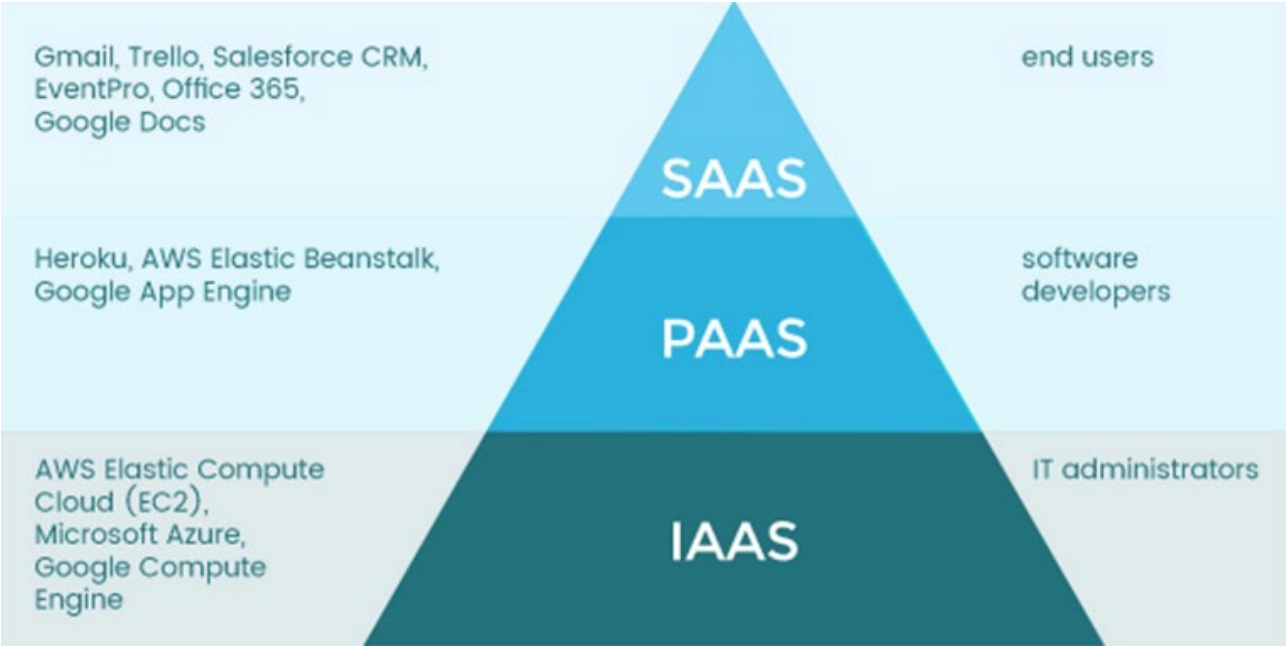


image address

IaaS vs. PaaS vs. SaaS

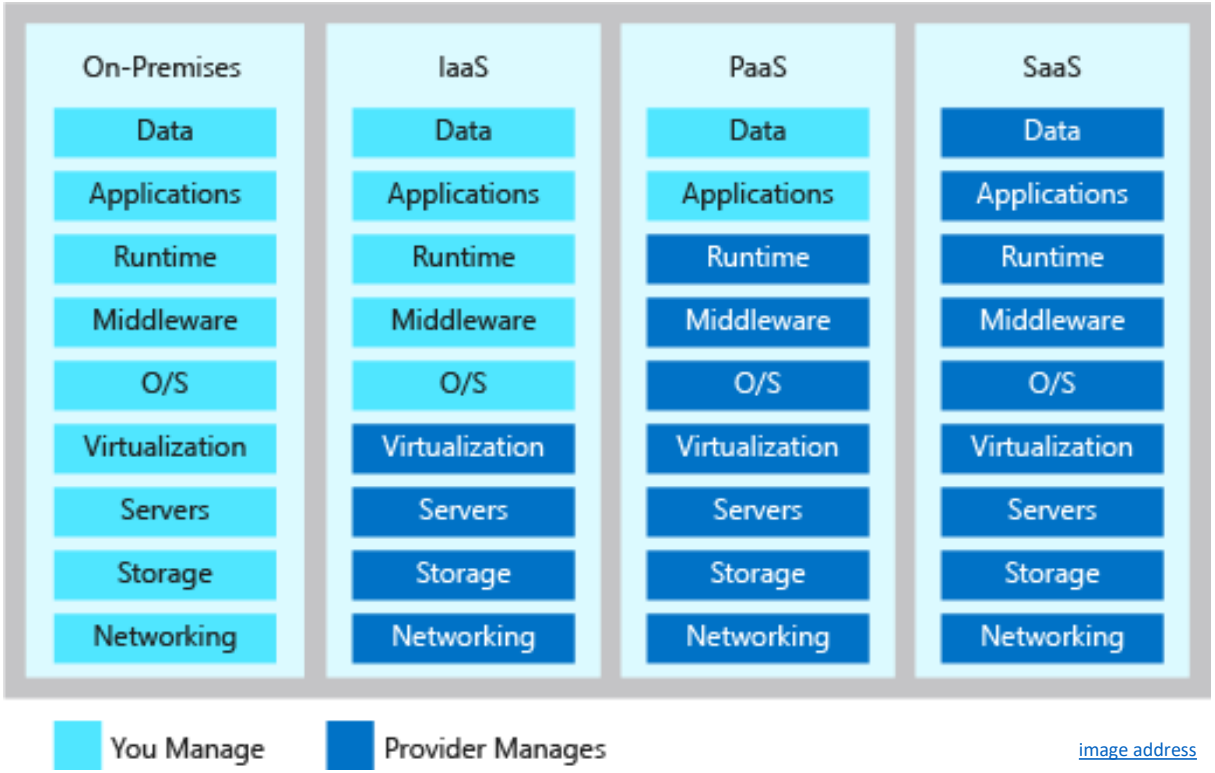
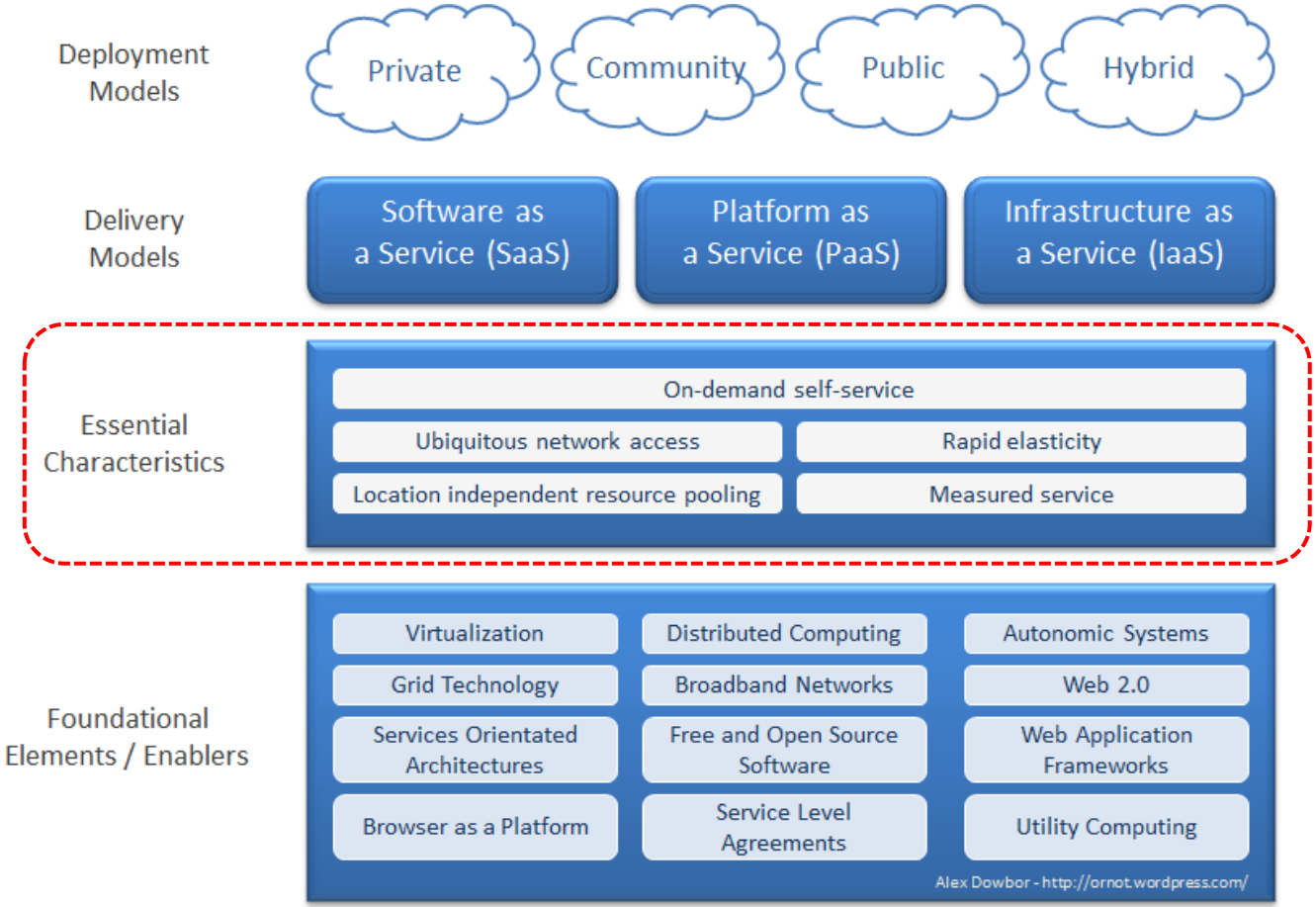


image address



Based on the NIST Working Definition of Cloud Computing v14 and http://www.csrc.nist.gov/groups/SNS/cloud-computing/index.html

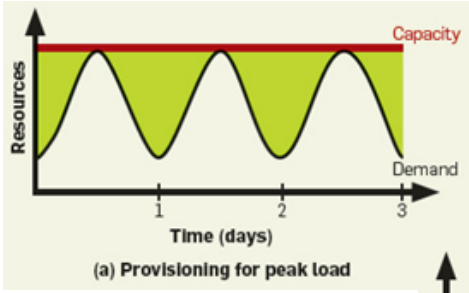
Creative Commons Attribution-Share Alike 3.0
Alexander Dowbor – http://ornot.wordpress.com

Cloud Computing Characteristics

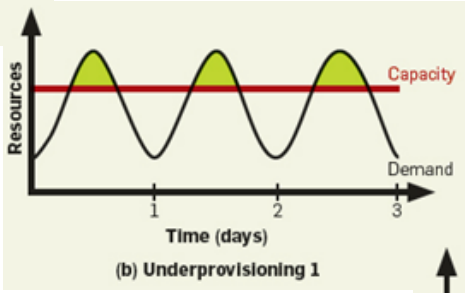
- On-demand self-service
  - The computing resources you need with self-service access
- Broad network access
  - Available over a network that can be accessed by diverse customer platforms
- Multi-tenancy & resource pooling
  - Multiple customers serviced from the provider's resource pool
- Rapid scalability & elasticity
  - Scale up or down quickly and easily to meet demand
- Measured/Metered service
  - Pay-for-use: You only pay for what you use

https://csrc.nist.gov/publications/detail/sp/800-145/final

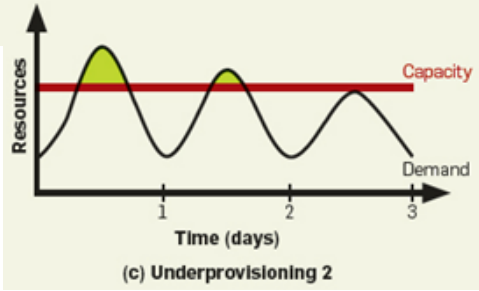
Static Service Provisioning



(a) Provisioning for peak load



(b) Underprovisioning 1

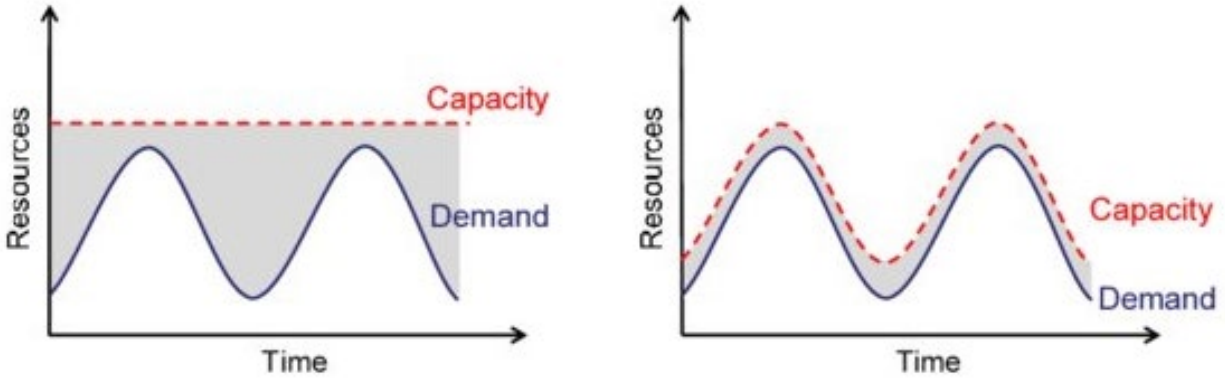


(c) Underprovisioning 2

(Armbrust et al., 2010)

Scalable Service Provisioning

- To avoid under-provisioning and over-provisioning
- To realise the cost model of 'pay-by-use'



(Armbrust et al., 2010)

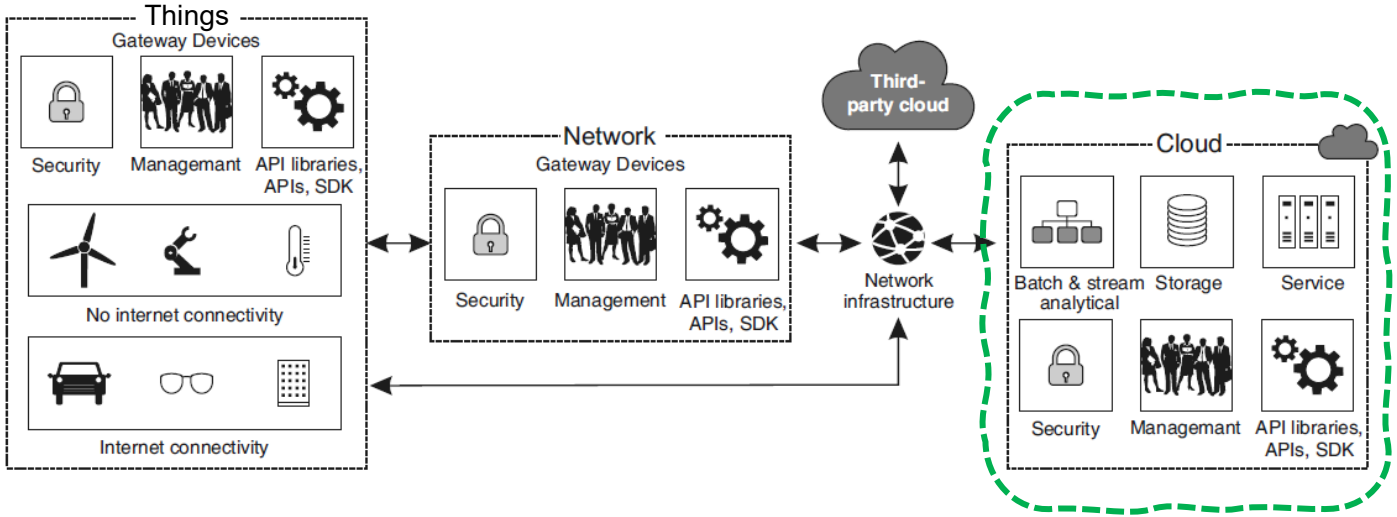
Cloud Computing: Benefits

- Cost
  - Eliminate the expenses of setting up and running on-site datacentres
- Speed
  - Provide self-service and on demand; resources provision in minutes
- Scalability (scale elastically)
  - Deliver the right amount of IT resources according to the need
- Productivity
  - Remove the need for “racking and stacking” tasks
- Performance
  - Regularly upgraded to the latest generation of computing hardware
- Reliability
  - Support data backup, disaster recovery and business continuity

Limitations of Cloud Computing	**Requirements** of IoT Applications
• **Connectivity to the cloud** is a pre-requisite	• Some IoT systems need to work even if connection is temporarily unavailable
• Assumes there is always **enough bandwidth** to collect data	• An overly strong assumption for IoT applications
• **Centralizes analytics** thus defining the lower bound reaction time	• Some IoT applications won't be able to wait for the data to get to the cloud, be analysed and for insights to get back
• **Security** measures	• IoT applications require better encryption mechanisms than the existing ones in cloud computing for data protection.

Intel IoT Platform

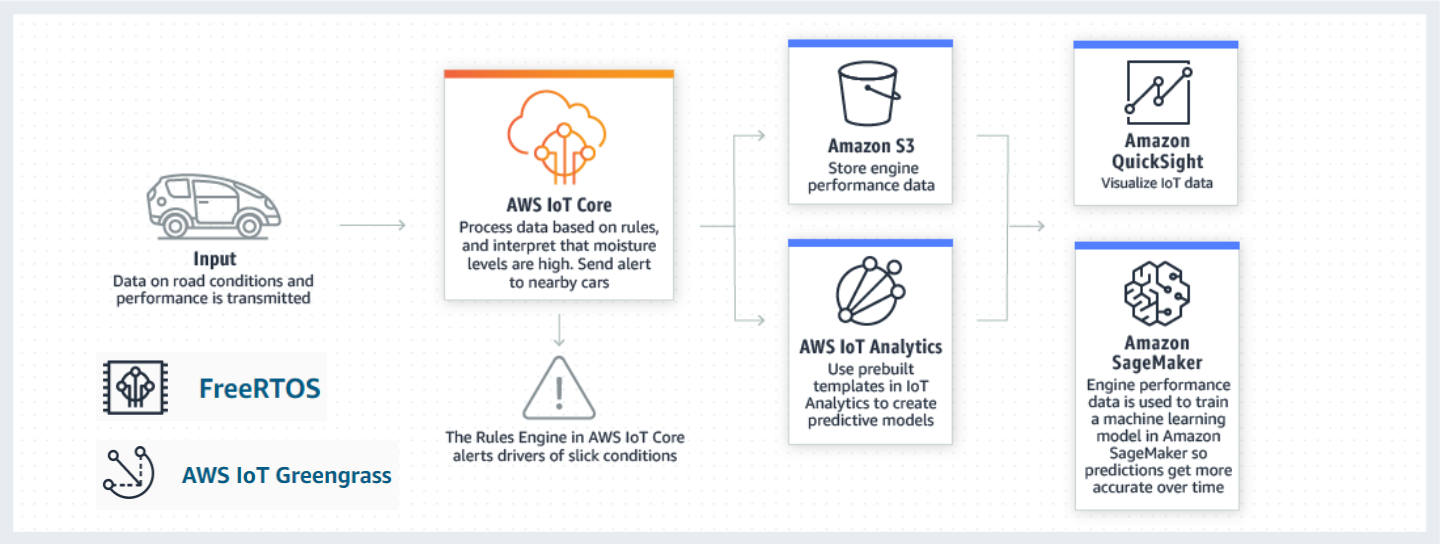
- Goal: To connect any device of any type to the Cloud
- A guideline for all parties involved



(Vuppalapati, 2020)

AWS IoT

- Bring together data management and rich analytics in easy-to-use services designed for noisy IoT data

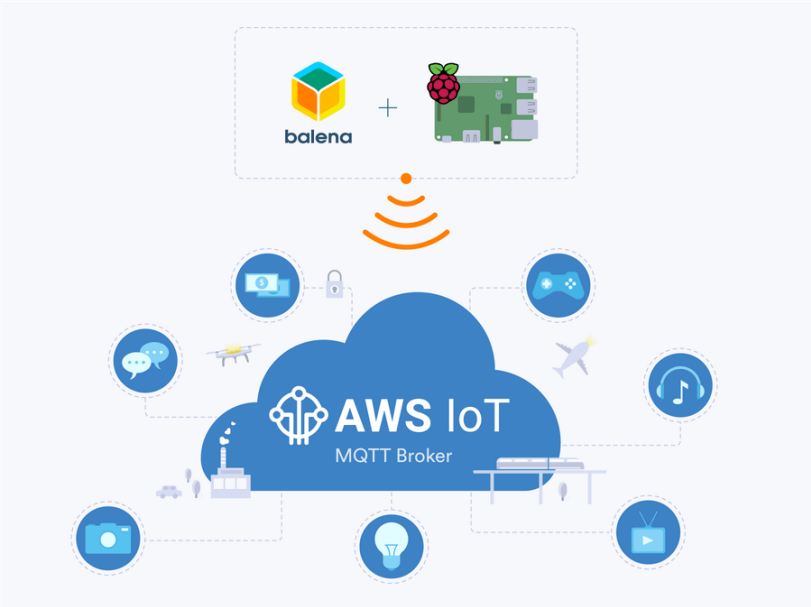


https://aws.amazon.com/iot/

https://docs.aws.amazon.com/whitepapers/latest/aws-overview/internet-of-things-services.html

AWS IoT: A practical project

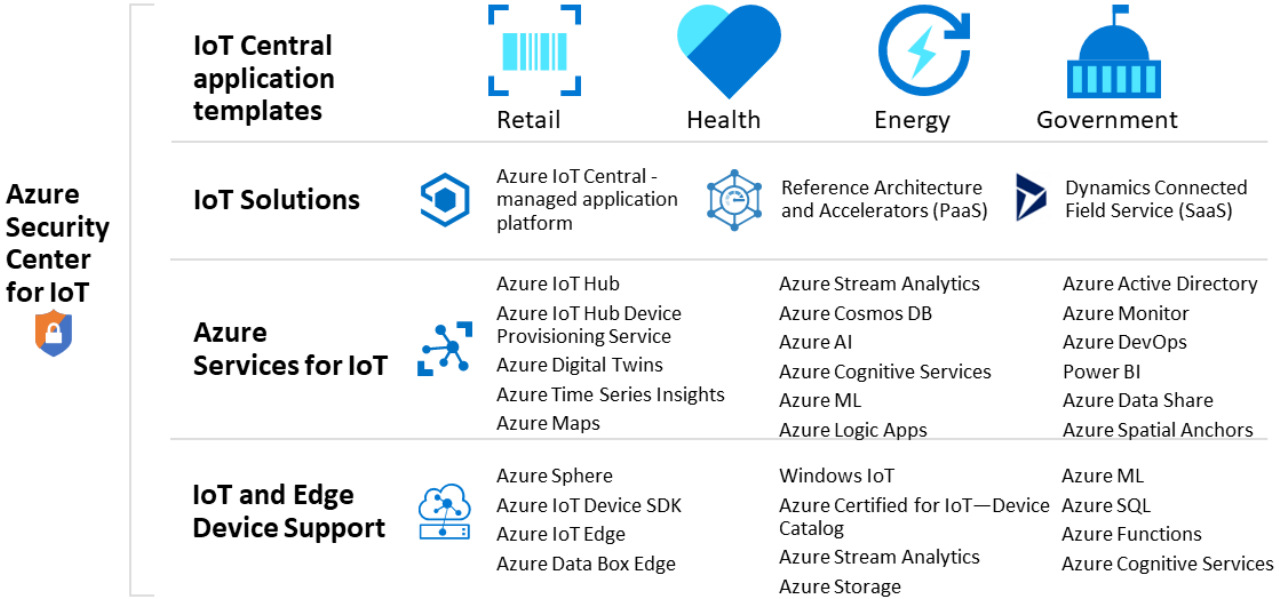
- Use a Raspberry Pi, via BalenaCloud, to Communicate with Amazon AWS IoT



https://www.hackster.io/dansku/use-a-raspberry-pi-to-communicate-with-amazon-aws-iot-e37525

Azure IoT

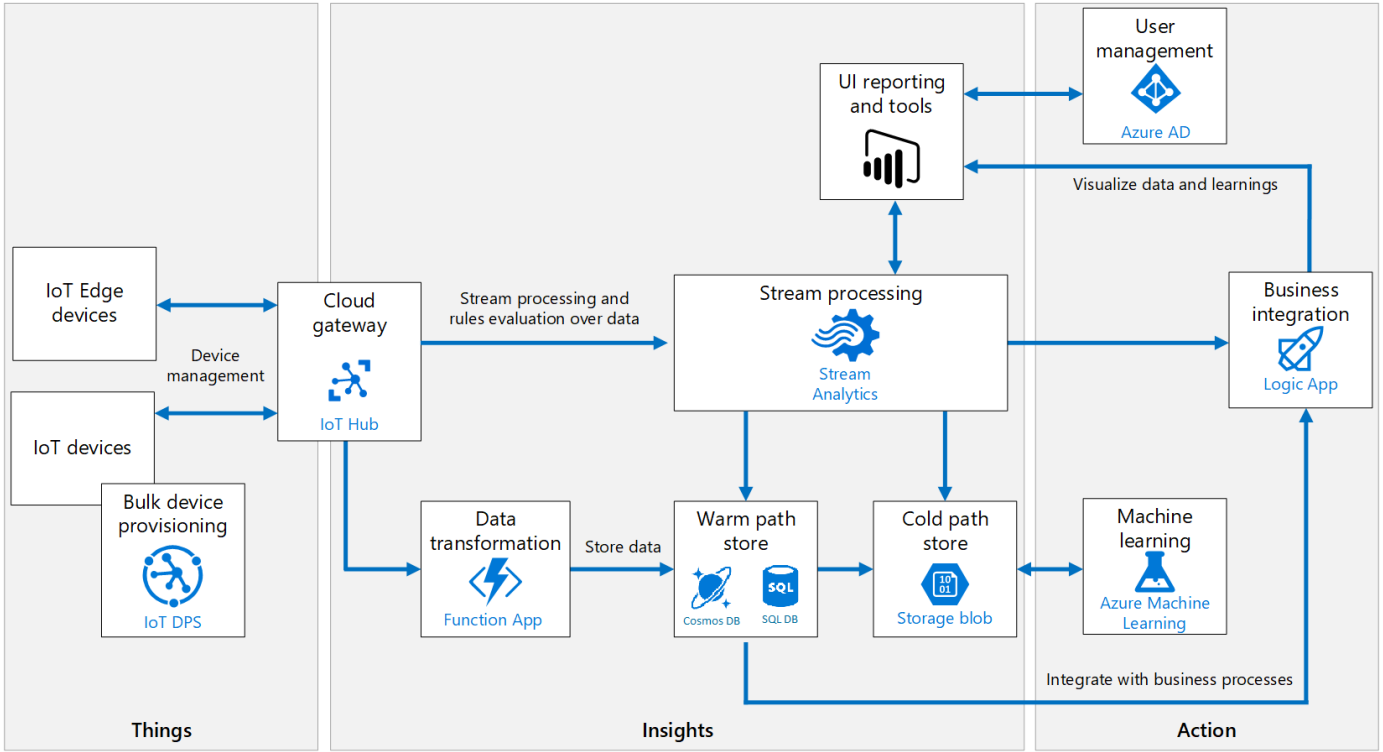
- A collection of Microsoft-managed cloud services that connect, monitor, and control billions of IoT assets



https://azure.microsoft.com/en-au/services/iot-hub/

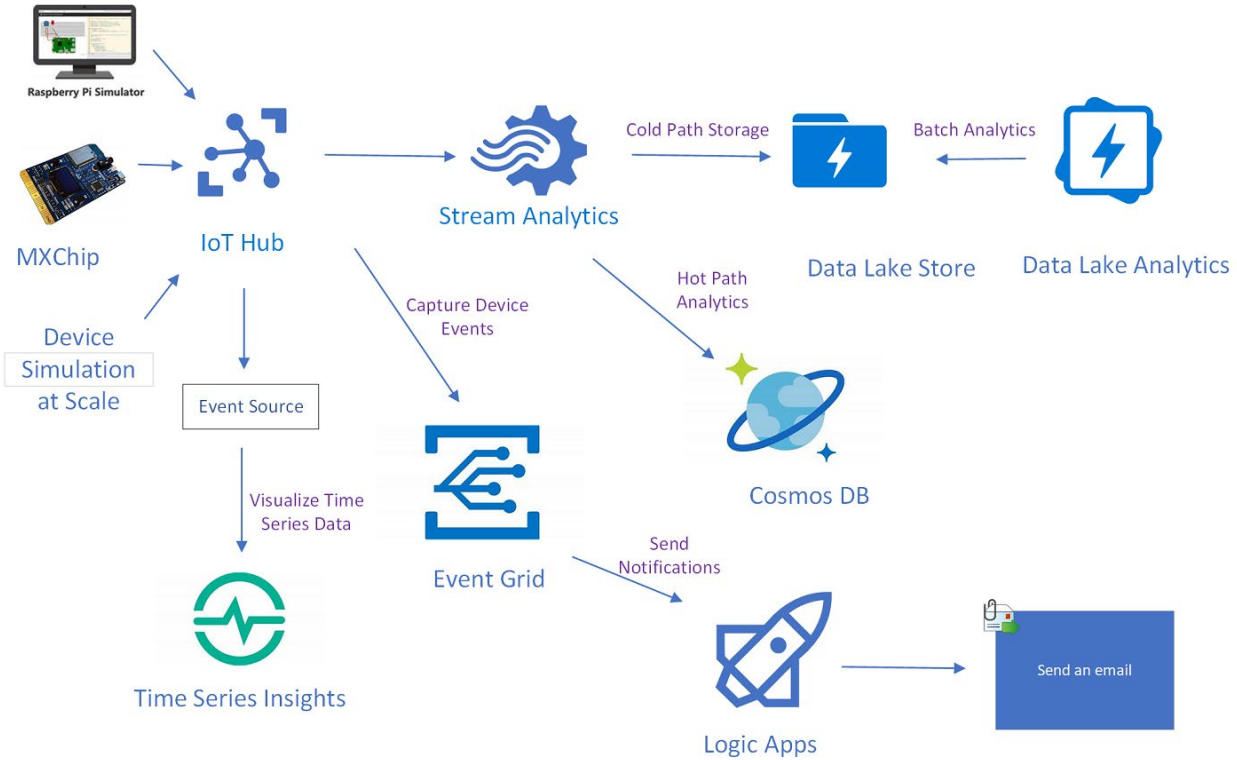
https://docs.microsoft.com/en-us/azure/iot-fundamentals/

Azure IoT – Reference Architecture



https://docs.microsoft.com/en-us/azure/architecture/reference-architectures/iot

Azure IoT: Hands on Labs



https://azure-samples.github.io/azureiotlabs/

Fog Computing

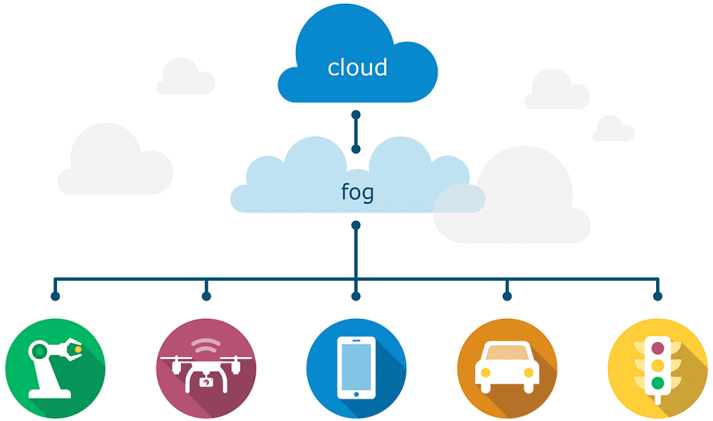


image address

What is Fog Computing?

- **Fog computing** (a.k.a. Edge computing) extends Cloud computing and services to the edge of the network

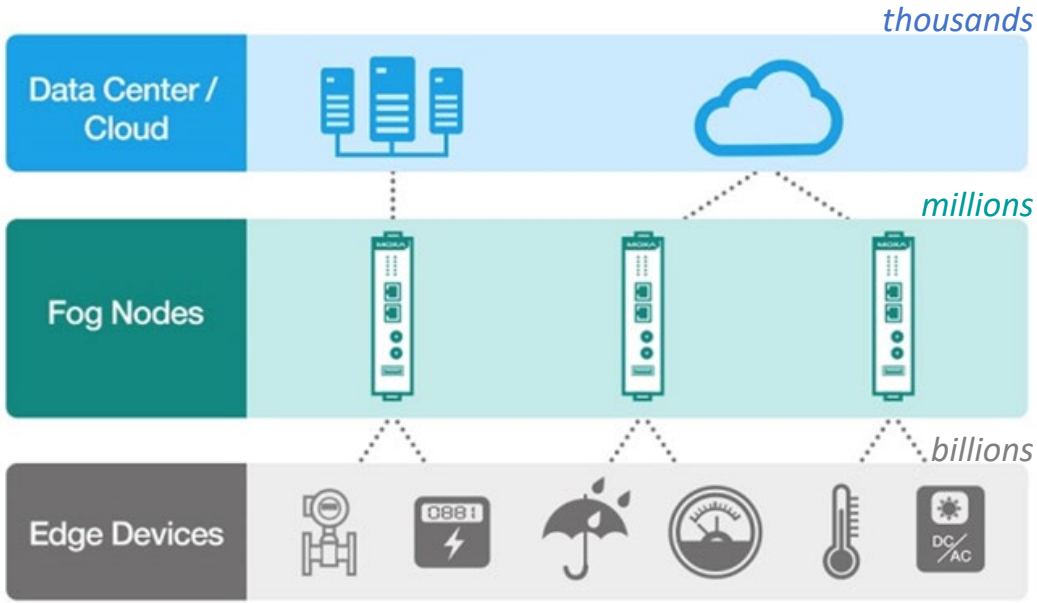
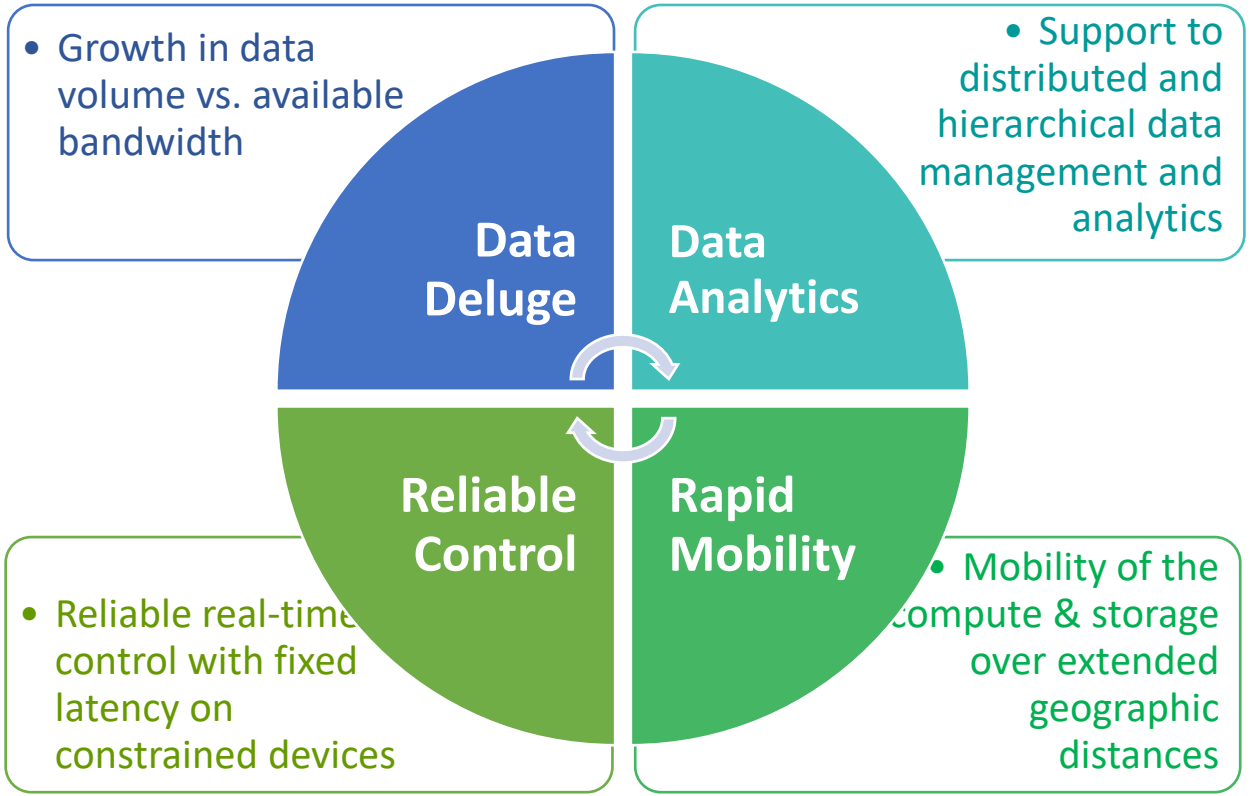


image address

Why Fog Computing?

- To improve efficiency and reduce the amount of data transported to the cloud for processing, analysis and storage
- To serve as a new model to ease wireless data transfer to distributed devices in the IoT applications
- Popular fog computing applications
  - Smart grid, smart city, smart buildings, vehicle networks, etc.
- "Cisco Fog Computing" is a registered name by Cisco, while "*fog computing*" is open to the community at large

Drivers for Fog Computing



(Rayes and Salam, 2022)

How Fog Computing Works?

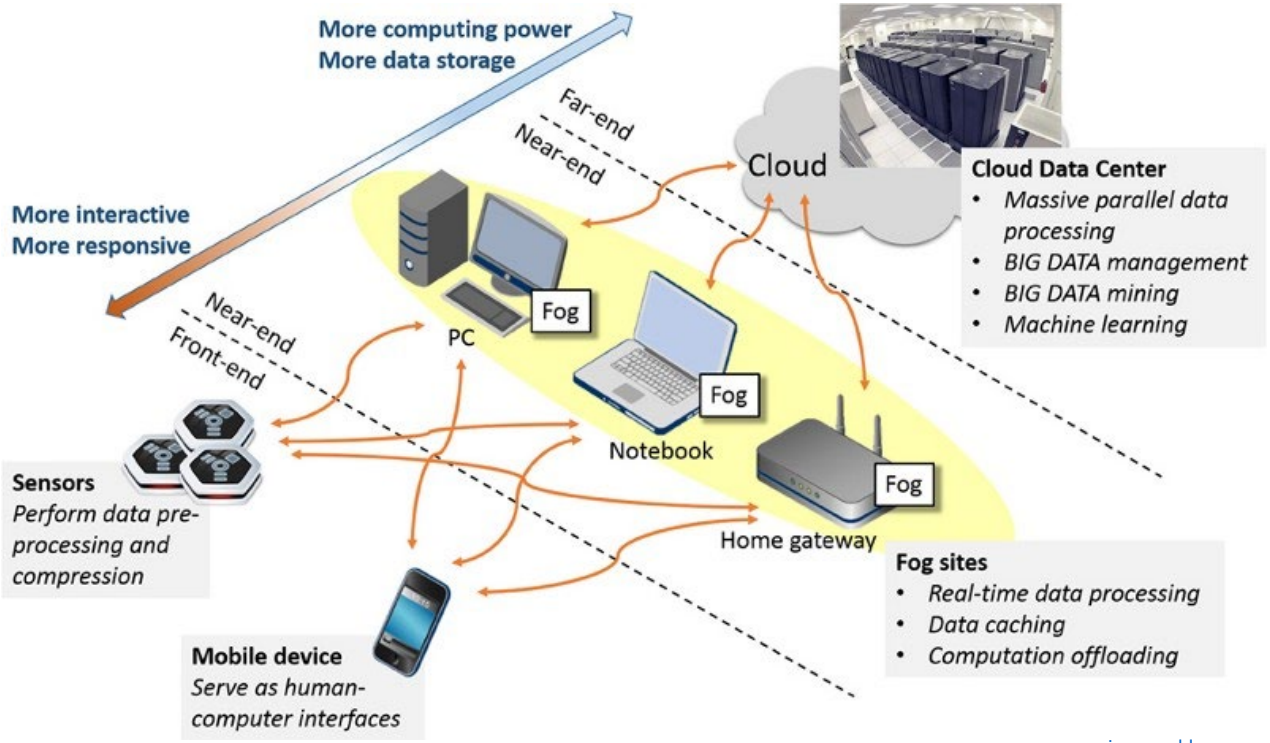
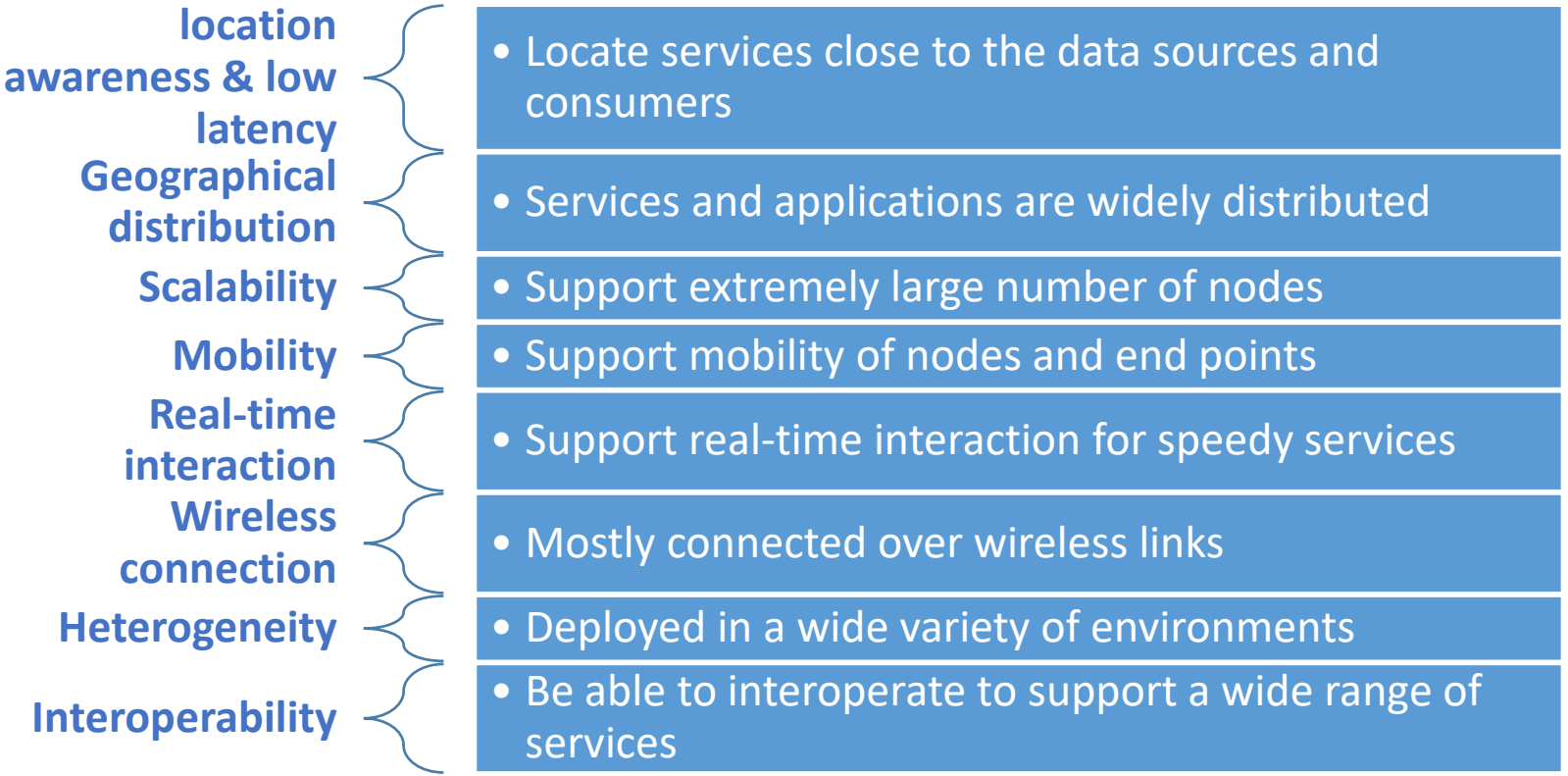


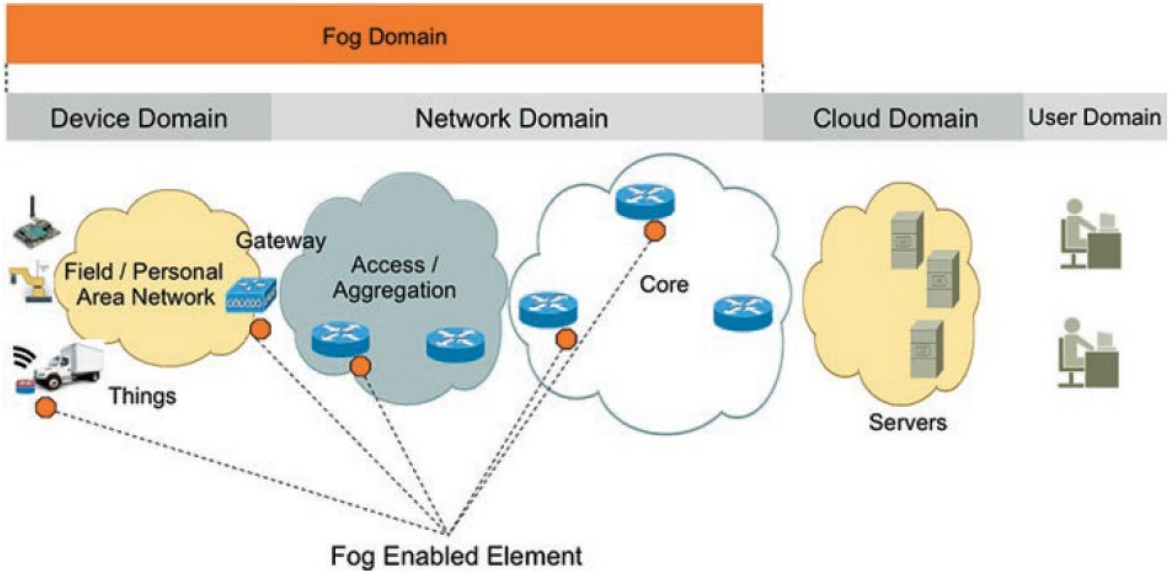
image address

Fog Computing Characteristics



Fog Computing in IoT

- Fog augments Cloud computing by bringing its functions closer to where data is produced or consumed in IoT applications
- Fog computing platform, located at the network edge, extends from IoT end devices and gateways to Cloud data centres



(Rayes and Salam, 2019)

How Fog Works for IoT?

- Developers write IoT applications for fog nodes at network edge
- Fog nodes closest to IoT devices ingest data from IoT devices
- Fog (aggregation) nodes then direct different types of data to the optimal place for analysis

	Fog Nodes Closest to IoT Devices	Fog Aggregation Nodes	Cloud
Response time	milliseconds to subsecond	seconds to minutes	minutes, days, weeks
Application examples	M2M communication, telemedicine	visualization, simple analytics	big data analytics, graphical dashboards
How long IoT data is stored	transient	Short duration: hours, days, weeks	months or years
Geographic coverage	local	wider	global

What Happens in the Fog and the Cloud?

Fog nodes

- Receive feeds from IoT devices using any protocol, in real time
- Run IoT-enabled applications for rea-time control and analytics, with millisecond response time
- Provide transient storage, often 1-2 hours
- Send periodic data summaries to the cloud

Cloud platform

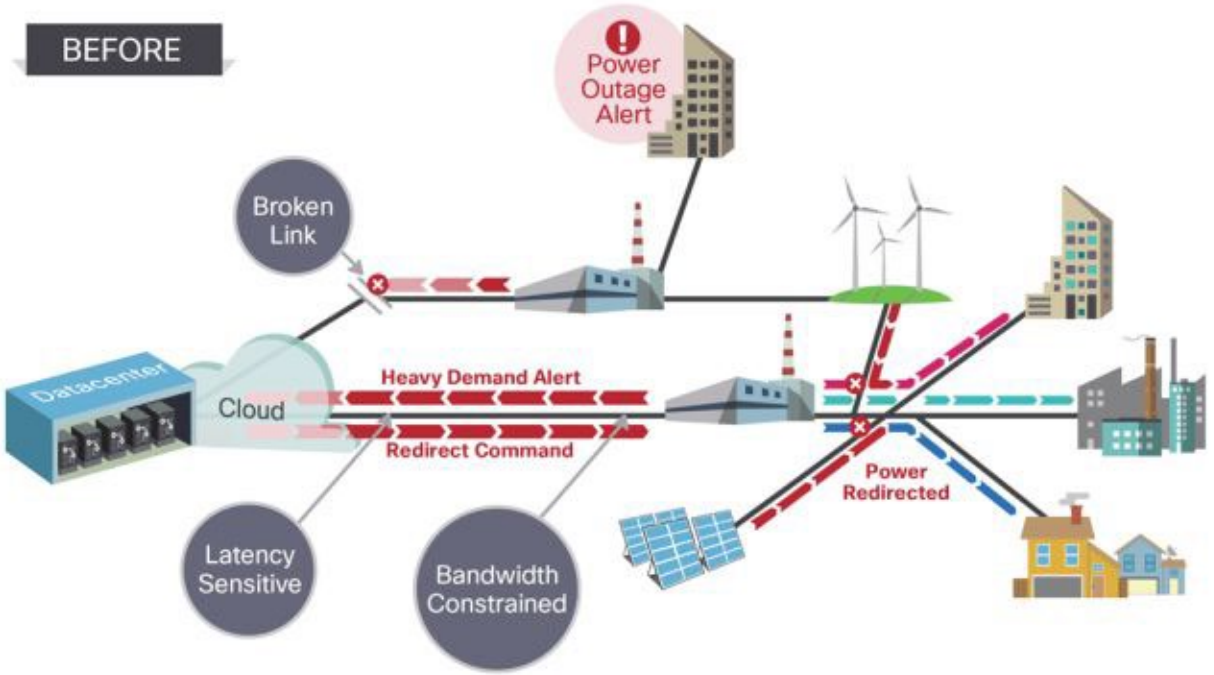
- Receives and aggregates data summaries from many fog nodes
- Performs analysis on the IoT data and data from other sources to gain business insights
- Can send new application rules to the fog nodes based on these insights

Fog vs. Cloud Computing

Requirement	Cloud computing	Fog computing
Latency	High	Low
Delay jitter	High	Very low
Location of service	Within Internet	Network edge
Distance between data sources and consumers	Multiple hops	Single hop
Location awareness	No	Yes
Geo-distribution	Centralised (data centre)	Distributed
# of server nodes	Medium – Large (homogeneous)	Large – Very Large (heterogeneous)
Support for mobility	Limited	Yes
Data analytics	Data at rest	Data in motion
Connectivity	Wired line	Wireless
Attack on data *en route*	High Probability	Less Probability

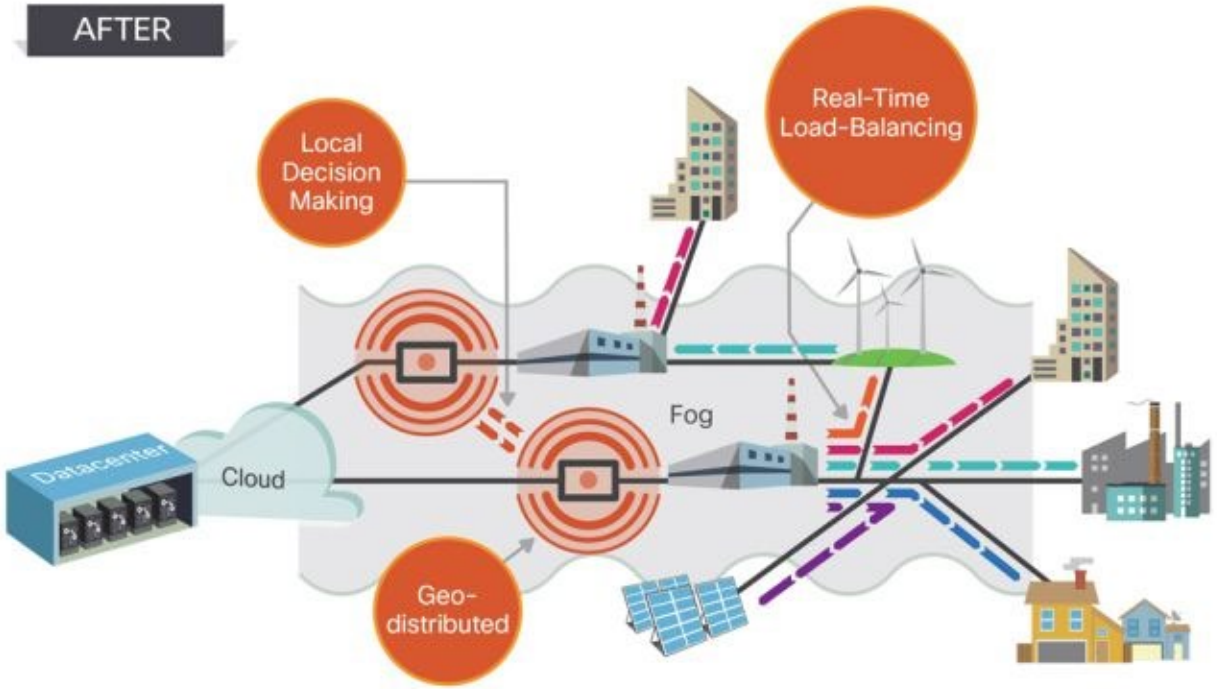
(Rayes and Salam, 2022)

Example: Smart Energy Distribution



https://sites.ndtv.com/digitizingindiaawards/fog-computing-collective-power-enables-a-new-breed-of-applications-and-services/

Example: Smart Energy Distribution



https://sites.ndtv.com/digitizingindiaawards/fog-computing-collective-power-enables-a-new-breed-of-applications-and-services/

OpenFog Reference Architecture



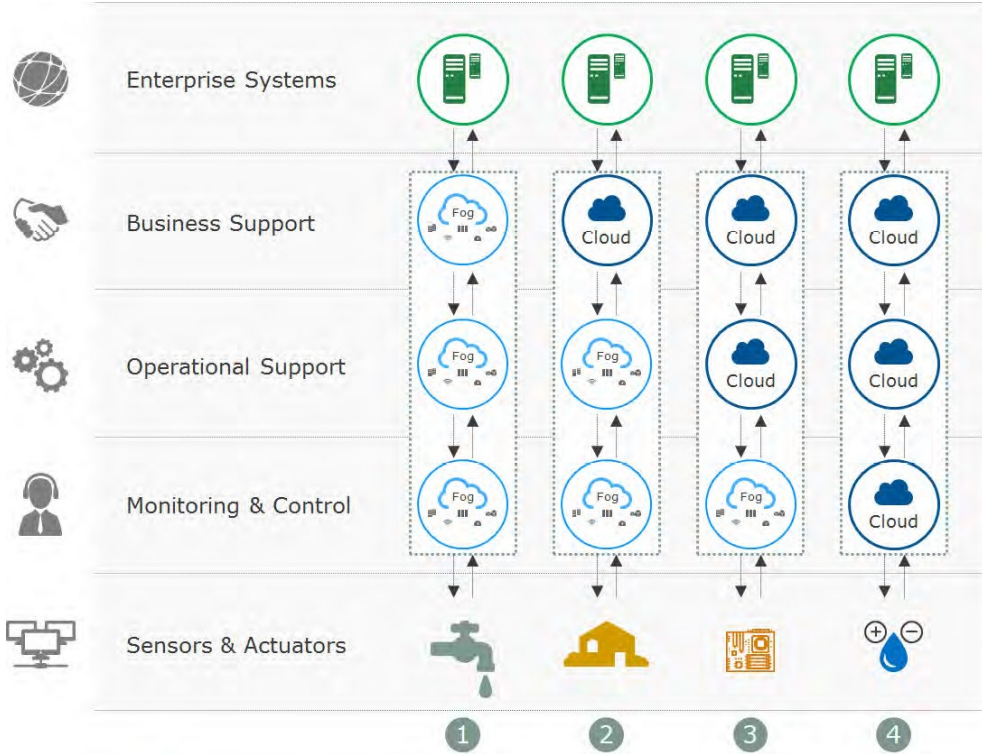
1934-2018 - IEEE Standard for Adoption of OpenFog Reference Architecture for Fog Computing | IEEE Standard | IEEE Xplore

OpenFog Reference Architecture (RA) – Pillars

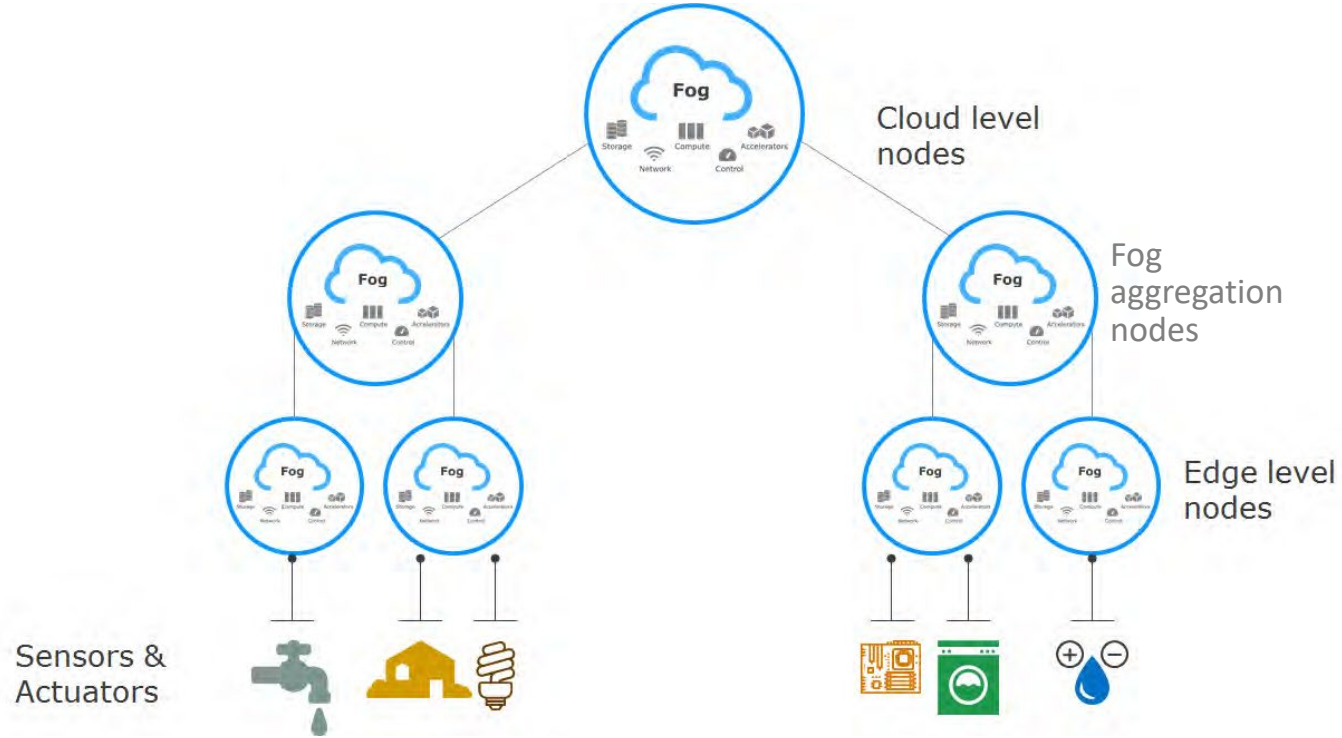
- Security
- Scalability
- Openness
- Autonomy
- Programmability
- RAS
  - Reliability
  - Availability
  - Serviceability
- Agility
- Hierarchy



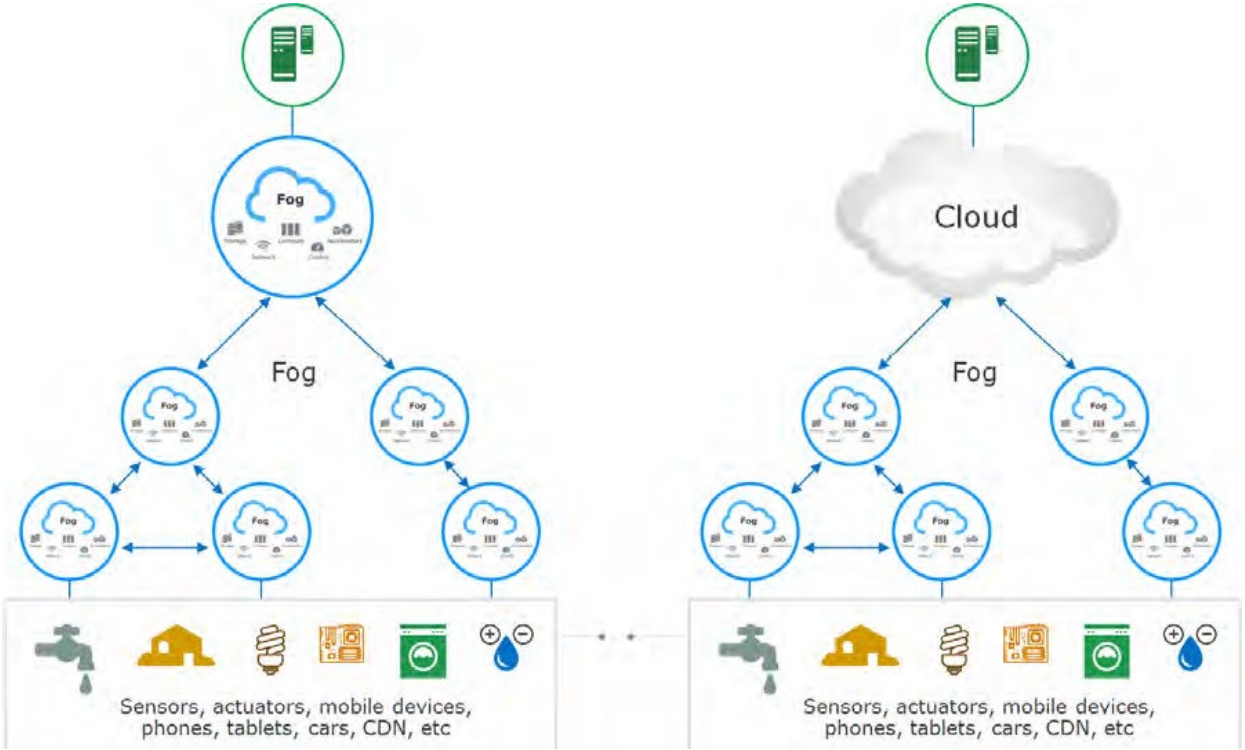
Fog Deployment Model



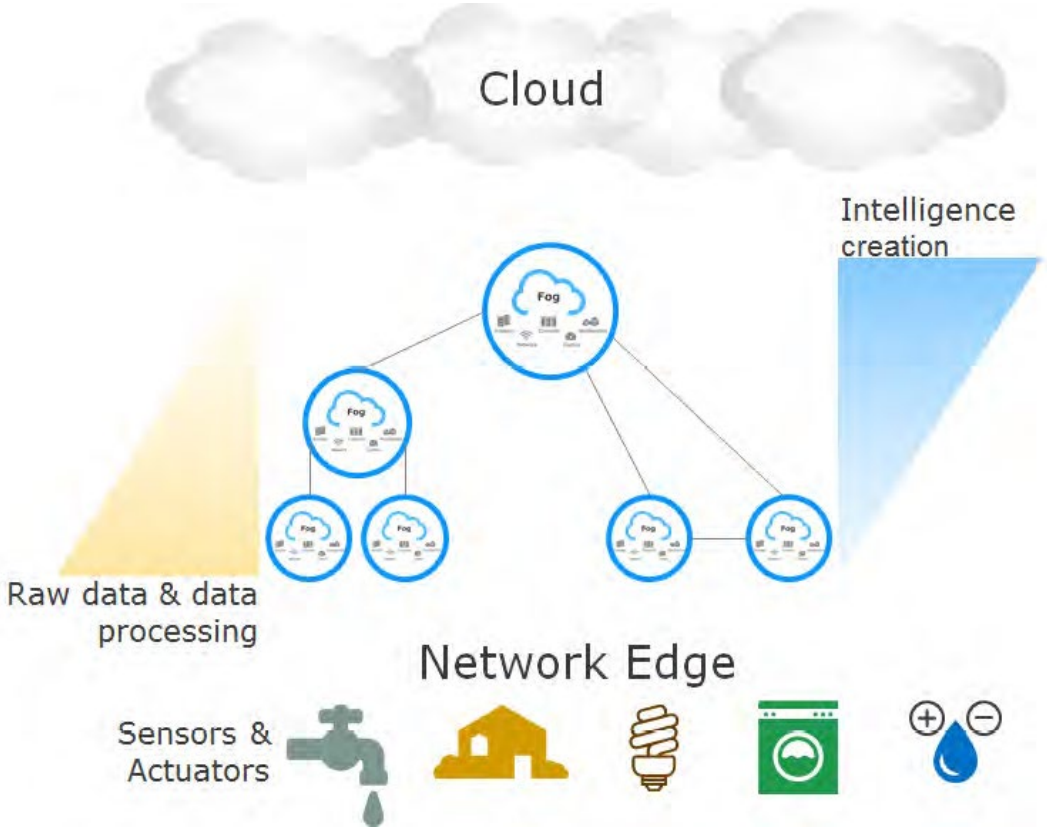
N-Tier Fog Deployment



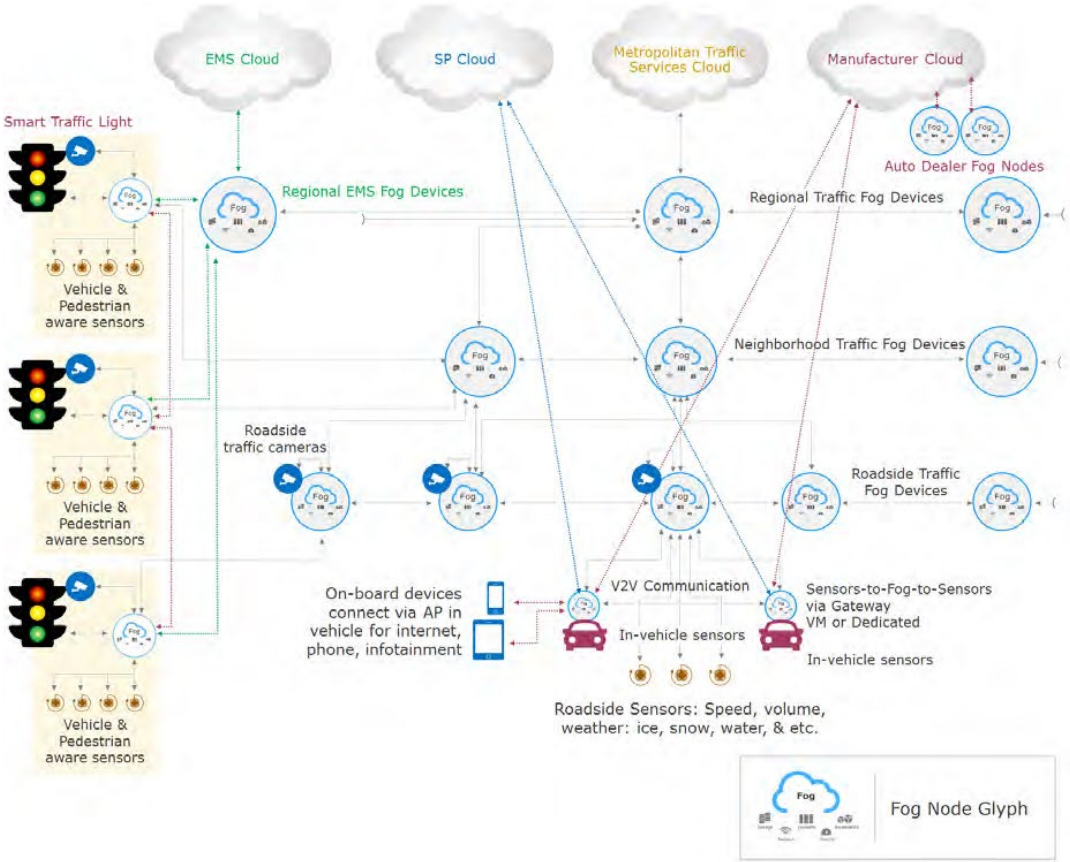
Cross-Fog Application



Data Processing Capability



Use Case 1:
Smart Car & Traffic Control System



Use Case 2:
Airport Visual Security

Passenger's journey

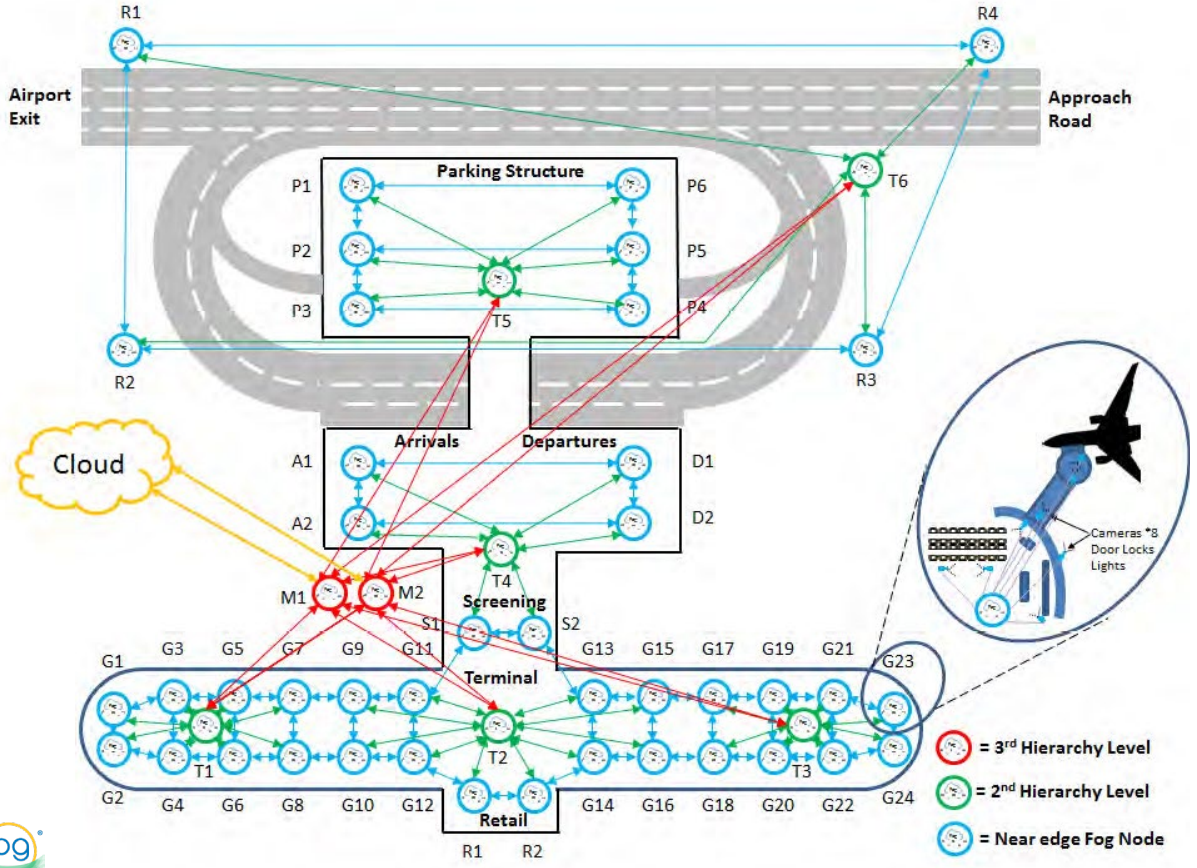
- Leaves from home and drives to the airport
- Parks in the long-term parking garage
- Takes bags to airport security checkpoint
- Bags are scanned and checked in
- Checks in through security and proceeds to boarding gate
- Upon arrival: Retrieves bags, proceeds to rental car agency, and leaves airport



Possible incidents/threats

- The vehicle entering the airport is stolen
- The passenger's name is on a no-fly list
- The passenger leaves his/her luggage someplace in the airport
- The passenger's luggage doesn't arrive with the flight
- The luggage is scanned and loaded on the plane, but it is not picked up by the correct passenger.
- An imposter steals (or switches) a boarding pass with another passenger and gets on someone else's flight.
- The passenger takes someone else's luggage at the arrival terminal

A Multi-Tier Solution Overview





Questions?